Space Mission Analytics User Manual

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1. Project Overview

Space Mission Analytics is an interactive web application built using Python, Streamlit, HTML, CSS, JavaScript, Three.js and Flask. It provides mission analytics, interactive dashboards, a 3D Solar System and AI-powered planet information.

2. Technologies Used

Python, Streamlit, Pandas, NumPy, Plotly, Matplotlib, HTML, CSS, JavaScript, Three.js, Flask and Google Gemini API.

3. Installation

Install Python 3.11 or later, then run:

pip install -r requirements.txt

4. Running the Project

Terminal 1
streamlit run xyzspace.py
Open: http://localhost:8501

Terminal 2
python -m http.server 8000
Serves HTML, CSS, JavaScript, images, videos and 3D assets.

Terminal 3
python app.py
Runs the Flask backend at http://127.0.0.1:5000

5. Features

• Interactive dashboard
• Historical mission analytics
• Interactive Plotly charts
• 3D Solar System using Three.js
• Glassmorphism UI
• Gemini AI integration

6. Gemini API Setup

Configure a valid Gemini API key before starting the Flask backend. If the API quota is exhausted, AI features will not return responses until a valid key with available quota is configured.

7. Deployment Notes

For local development the project uses Streamlit, a Python HTTP server and a Flask backend. For cloud deployment, configure these services appropriately.

8. Python Packages

streamlit
pandas
numpy
plotly
matplotlib
openpyxl
flask
google-genai
requests
streamlit-option-menu

9. Developer

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